Exam Practice: Mixed Review

Year 5 Accelerated · **Lesson Worksheet**

Worked Examples

Four Operations & Fractions: Calculate 2/5 of 150, then add 18.

Answer: 2/5 of 150 = 60; 60 + 18 = 78.

Ratio & Percentages: A sum of £80 is shared in the ratio 3:5. Find 25% of the smaller share.

Answer: Smaller share = 3/8 × £80 = £30; 25% of £30 = £7.50.

Speed, Distance & Time: A cyclist travels 36 km in 2 hours 15 minutes. Find her average speed in km/h.

Answer: 2 h 15 min = 2.25 h; speed = 36 ÷ 2.25 = 16 km/h.

Mixed Core (15 questions)

1. Write the value of the digit 7 in the number 47,208.
2. Calculate 3,847 + 2,956.
3. Calculate 9,004 − 3,758.
4. What is 34 × 6? A) 194 B) 204 C) 214 D) 224 E) 234
5. Calculate 5,184 ÷ 8.
6. What is 3/8 of 64?
7. Calculate 7.65 + 4.8. A) 11.45 B) 12.45 C) 12.65 D) 13.45 E) 12.05
8. Find 15% of 240.
9. Share £60 in the ratio 2:3. What is the larger share?
10. If x + 9 = 23, what is the value of x? A) 12 B) 13 C) 14 D) 15 E) 16
11. How many degrees are there in a right angle?
12. Convert 2.5 kg into grams.
13. The mean of 4, 8, 6 and 10 is: A) 6 B) 6.5 C) 7 D) 7.5 E) 8
14. Select ALL that apply: which of these numbers are multiples of 9? A) 27 B) 42 C) 63 D) 80 E) 81
15. A car travels at a steady 40 mph for 2 hours. How far does it travel?

Mixed Applied (15 questions)

1. Tom buys 3 notebooks at £1.45 each and a pen for 85p. How much does he spend in total?
2. A recipe needs 250 g of flour to make 10 cookies. How much flour is needed for 24 cookies?
3. A train journey covers 180 miles at an average speed of 60 mph. How long does the journey take? A) 2 hours B) 2.5 hours C) 3 hours D) 3.5 hours E) 4 hours
4. In a class of 30 pupils, 40% walk to school. How many pupils walk to school?

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5. A rectangular garden is 8 m long and 5 m wide. What is its area?

6. Sarah has £25. She spends 2/5 of it on a book. How much money does she have left?

7. A shop reduces a £45 jacket by 20% in a sale. What is the sale price? A) £33 B) £34 C) £35 D) £36 E) £37

8. Three friends share a £36 restaurant bill equally. How much does each person pay?

9. A jug holds 1.8 litres of juice. 650 ml is poured out. How much juice remains, in millilitres?

10. Select ALL that apply: a bag contains red, blue and green counters in the ratio 3:2:5. Which statements are true? A) For every 10 counters, 3 are red. B) Green counters are the most common colour. C) Blue counters make up 1/5 of the bag. D) Red counters make up 3/10 of the bag. E) There are more blue counters than red counters.

11. A clock shows 10:40 am. What time will it show in 55 minutes?

12. A football match lasts 90 minutes plus 4 minutes of added time. If kick-off is at 3:00 pm, at what time does the match end (ignoring any half-time break)?

13. A number is tripled and then 7 is subtracted, giving 29. What was the original number? A) 10 B) 11 C) 12 D) 13 E) 14

14. A school hall measures 15 m by 12 m. Carpet tiles cost £8 per square metre. What is the total cost of carpeting the hall?

15. A car's fuel tank holds 50 litres. The car uses fuel at a rate of 8 litres per 100 km. How far can it travel on a full tank?

Mixed Challenge (15 questions)

1. A theatre sells tickets at £12 for adults and £7 for children. On Saturday it sells 85 adult tickets and 120 child tickets. What is the theatre's total ticket income?

2. A tank is 2/3 full of water. After 40 litres are removed, it is 2/5 full. What is the capacity of the tank?

3. A number is increased by 20% and then decreased by 20%, giving 96. What was the original number? A) 96 B) 98 C) 99 D) 100 E) 102

4. Ben, Chloe and Dan share £180 so that Ben gets twice as much as Chloe, and Dan gets £20 more than Chloe. How much does Dan receive?

5. A cyclist rides the first 15 km of a journey at 20 km/h and the remaining 10 km at 15 km/h. What is her average speed for the whole journey, to 1 decimal place?

6. Select ALL that apply: a sequence begins 4, 9, 16, 25, 36 … Which statements about this sequence are true? A) Each term is a square number. B) The next term is 49. C) The term-to-term rule is 'add 7 each time'. D) The 10th term in the sequence is 121. E) The differences between consecutive terms increase by 2 each time.

7. The exterior angle of a regular polygon is 24°. How many sides does the polygon have? A) 12 B) 13 C) 14 D) 15 E) 16

8. A map has a scale of 1 : 25,000. Two towns are 8.4 cm apart on the map. What is the actual distance between them, in kilometres?

9. A pattern of tiles uses 3 tiles for 1 row, 7 tiles for 2 rows, and 13 tiles for 3 rows. If the pattern continues in the same way, how many tiles are needed for 5 rows?

10. In a survey of 200 students, the ratio of those who prefer Maths to those who prefer English is 7:3. How many more students prefer Maths than English?

11. A square has the same perimeter as a rectangle measuring 18 cm by 12 cm. What is the area of the square? A) 196 cm² B) 210 cm² C) 225 cm² D) 240 cm² E) 256 cm²

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12. A water butt starts with 96 litres and leaks at a steady rate, losing 1/8 of its original contents every hour. How many litres remain after 3 hours?

13. Three consecutive even numbers add up to 90. What is the largest of the three numbers?

14. A shop buys a laptop for £450 and sells it at a profit of 28%. What is the selling price?

15. A rectangular swimming pool is 25 m long, 10 m wide and 2 m deep. It is filled using a pump that delivers 2,500 litres per minute. How long, in minutes, does it take to fill the pool completely? (1 m^3 = 1,000 litres)

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